

Vapor-Compression Refrigeration Cycle Analysis

CBE 20260 / Thermodynamics I

This interactive notebook solves a standard vapor-compression refrigeration cycle. You can select different working fluids (refrigerants) to see how their thermodynamic properties impact the coefficient of performance (COP) and the cooling load.

Assumptions: 1. The evaporator and condenser operate at constant pressure. 2. The refrigerant leaves the evaporator as a saturated vapor ($x = 1$). 3. The refrigerant leaves the condenser as a saturated liquid ($x = 0$). 4. The expansion valve is an isenthalpic throttle ($\Delta H = 0$). 5. The compressor has an adjustable isentropic efficiency (η_c).

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# LECTURE 22: Vapor-Compression Refrigeration Cycle Analysis
# Script: vapor-compression-cycle.ipynb
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#
# Description:
# This notebook performs a thermodynamic analysis of a standard
#   ↪ vapor-compression
# refrigeration cycle. It calculates the states, work,
# heats, and COP of the cycle based on user inputs for the
#   ↪ pressure of the
# evaporator and condenser (or the temperatures, depending on
#   ↪ the mode selected).
# The efficiency of the compressor is specified by the user.
# The notebook uses CoolProp to obtain fluid properties and
#   ↪ ipywidgets to create
# an interactive interface for exploring how different operating
#   ↪ conditions and
# refrigerants affect cycle performance.
#
# Note that the critical point of the refrigerant is a hard
#   ↪ limit for the cycle.
# If you exceed it, CoolProp will throw an error, which is
#   ↪ caught and displayed
# as a user-friendly message.
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# Install CoolProp if it is missing (e.g., on Google Colab)
try:
    import CoolProp
except ImportError:
    %pip install -q CoolProp
import CoolProp.CoolProp as CP
import ipywidgets as widgets
from IPython.display import display, clear_output

def solve_vcc(fluid, mode, T_evap_C, T_cond_C, P_evap_bar,
↪ P_cond_bar, eta):
    """
    Calculates the states and performance of a standard
    ↪ vapor-compression
    refrigeration cycle.
    """
    try:
        # --- 1. Determine Pressures and Temperatures ---
        if mode == 'Temperatures (°C)':
            T_evap = T_evap_C + 273.15
            T_cond = T_cond_C + 273.15

            if T_evap >= T_cond:
                print("Error: Evaporator temperature must be
                ↪ lower than Condenser temperature.")
                return

            # Find saturation pressures
            P_evap = CP.PropsSI('P', 'T', T_evap, 'Q', 1, fluid)
            P_cond = CP.PropsSI('P', 'T', T_cond, 'Q', 0, fluid)

        else:
            P_evap = P_evap_bar * 1e5
            P_cond = P_cond_bar * 1e5

            if P_evap >= P_cond:
                print("Error: Evaporator pressure must be lower
                ↪ than Condenser pressure.")
                return

            # Find saturation temperatures
            T_evap = CP.PropsSI('T', 'P', P_evap, 'Q', 1, fluid)
            T_cond = CP.PropsSI('T', 'P', P_cond, 'Q', 0, fluid)

        # --- 2. Calculate State 1 (Evaporator Out / Compressor
        ↪ In) ---

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# Saturated Vapor
H1 = CP.PropsSI('H', 'P', P_evap, 'Q', 1, fluid) / 1000
↪ # kJ/kg
S1 = CP.PropsSI('S', 'P', P_evap, 'Q', 1, fluid) / 1000
↪ # kJ/kg-K

# --- 3. Calculate State 2 (Compressor Out / Condenser
↪ In) ---
# Ideal Isentropic Compression
H2_ideal = CP.PropsSI('H', 'P', P_cond, 'S', S1 * 1000,
↪ fluid) / 1000
W_c_ideal = H2_ideal - H1

# Actual Compression
W_c = W_c_ideal / eta
H2 = H1 + W_c
T2 = CP.PropsSI('T', 'P', P_cond, 'H', H2 * 1000, fluid)

# --- 4. Calculate State 3 (Condenser Out / Valve In)
↪ ---
# Saturated Liquid
H3 = CP.PropsSI('H', 'P', P_cond, 'Q', 0, fluid) / 1000

# --- 5. Calculate State 4 (Valve Out / Evaporator In)
↪ ---
# Isenthalpic Throttling
H4 = H3
x4 = CP.PropsSI('Q', 'P', P_evap, 'H', H4 * 1000, fluid)

# --- 6. Cycle Performance Calculations ---
q_in = H1 - H4 # Cooling Load (Heat absorbed in
↪ Evaporator)
q_out = H3 - H2 # Heat rejected in Condenser
↪ (IUPAC: negative)
COP = q_in / W_c # Coefficient of Performance

# --- 7. Print Results ---
print("=" * 60)
print(f"VAPOR-COMPRESSION CYCLE ANALYSIS: {fluid}")
print("=" * 60)
print(f"Evaporator: {P_evap/1e5:.2f} bar |
↪ {T_evap-273.15:.2f} °C")
print(f"Condenser: {P_cond/1e5:.2f} bar |
↪ {T_cond-273.15:.2f} °C")
print("-" * 60)
print(f"State 1 (Comp In) : H1 = {H1:>6.2f} kJ/kg | S1
↪ = {S1:.4f} kJ/kg-K")

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    print(f"State 2 (Comp Out) : H2 = {H2:>6.2f} kJ/kg | T2
    ↪ = {T2-273.15:.2f} °C")
    print(f"State 3 (Valve In) : H3 = {H3:>6.2f} kJ/kg |
    ↪ Saturated Liquid")
    print(f"State 4 (Valve Out): H4 = {H4:>6.2f} kJ/kg |
    ↪ Quality (x) = {x4:.3f}")
    print("-" * 60)
    print(f"Cooling Load (q_in):      {q_in:.2f} kJ/kg")
    print(f"Compressor Work (W_c):     {W_c:.2f} kJ/kg")
    print(f"Heat Rejected (q_out):     {q_out:.2f} kJ/kg")
    print("-" * 60)
    print(f"COEFFICIENT OF PERFORMANCE (COP): {COP:.2f}")
    print("=" * 60)

except ValueError as e:
    print("\n--- FLUID PROPERTY ERROR ---")
    print("The selected conditions are invalid for this
    ↪ fluid.")
    print("You may have exceeded the critical point (e.g.,
    ↪ condensing CO2 above ~31°C).")
    print(f"CoolProp Error: {e}")
    print("Try adjusting the temperatures/pressures or
    ↪ selecting a different fluid.")

# --- CREATE INTERACTIVE WIDGETS ---
style = {'description_width': 'initial'}

# Fluid & Mode Selection
dropdown_fluid = widgets Dropdown(options=['R134a', 'Ammonia',
    ↪ 'CO2', 'R22', 'R410A', 'IsoButane'], value='R134a',
    ↪ description='Refrigerant:', style=style)
dropdown_mode = widgets Dropdown(options=['Temperatures (°C)',
    ↪ 'Pressures (bar)'], value='Temperatures (°C)',
    ↪ description='Input Mode:', style=style)

# Temperature Sliders
slider_T_evap = widgets.FloatSlider(value=-10.0, min=-40.0,
    ↪ max=15.0, step=1.0, description='Evap Temp (°C):',
    ↪ style=style)
slider_T_cond = widgets.FloatSlider(value=30.0, min=10.0,
    ↪ max=60.0, step=1.0, description='Cond Temp (°C):',
    ↪ style=style)
box_T = widgets.VBox([slider_T_evap, slider_T_cond])

# Pressure Sliders (Hidden by default)
slider_P_evap = widgets.FloatSlider(value=2.0, min=0.5,
    ↪ max=15.0, step=0.5, description='Evap Pressure (bar):',
    ↪ style=style)

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slider_P_cond = widgets.FloatSlider(value=10.0, min=5.0,
    ↪ max=60.0, step=0.5, description='Cond Pressure (bar):',
    ↪ style=style)
box_P = widgets.VBox([slider_P_evap, slider_P_cond])
box_P.layout.display = 'none' # Hide initially

slider_eta = widgets.FloatSlider(value=0.75, min=0.6, max=1.0,
    ↪ step=0.01, description='Compressor Efficiency ( $\eta$ ):',
    ↪ style=style)

# Dynamic UI toggling based on Mode selection
def on_mode_change(change):
    if change['new'] == 'Temperatures (°C)':
        box_T.layout.display = 'flex'
        box_P.layout.display = 'none'
    else:
        box_T.layout.display = 'none'
        box_P.layout.display = 'flex'

dropdown_mode.observe(on_mode_change, names='value')

# Group everything
ui = widgets.VBox([dropdown_fluid, dropdown_mode, box_T, box_P,
    ↪ slider_eta])

# Connect to function
out = widgets.interactive_output(solve_vcc, {
    'fluid': dropdown_fluid,
    'mode': dropdown_mode,
    'T_evap_C': slider_T_evap,
    'T_cond_C': slider_T_cond,
    'P_evap_bar': slider_P_evap,
    'P_cond_bar': slider_P_cond,
    'eta': slider_eta
})

# Display
display(ui, out)

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VBox(children=(Dropdown(description='Refrigerant:', options=('R134a', 'Ammonia'), value='R134a'),

Output())

```

# Static preview: results for the default inputs above.
# The output is saved so the course website can show it.
# To try other inputs, use the controls above, or open

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# this notebook in Colab (badge on the course website).
solve_vcc(fluid=dropdown_fluid.value, mode=dropdown_mode.value,
          T_evap_C=slider_T_evap.value,
          ↪ T_cond_C=slider_T_cond.value,
            P_evap_bar=slider_P_evap.value,
          ↪ P_cond_bar=slider_P_cond.value,
            eta=slider_eta.value)
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VAPOR-COMPRESSION CYCLE ANALYSIS: R134a
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Evaporator: 2.01 bar | -10.00 °C
Condenser: 7.70 bar | 30.00 °C
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State 1 (Comp In) : H1 = 392.66 kJ/kg | S1 = 1.7334 kJ/kg-K
State 2 (Comp Out) : H2 = 429.89 kJ/kg | T2 = 44.56 °C
State 3 (Valve In) : H3 = 241.72 kJ/kg | Saturated Liquid
State 4 (Valve Out): H4 = 241.72 kJ/kg | Quality (x) = 0.267
-----
Cooling Load (q_in): 150.94 kJ/kg
Compressor Work (W_c): 37.23 kJ/kg
Heat Rejected (q_out): -188.17 kJ/kg
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COEFFICIENT OF PERFORMANCE (COP): 4.05
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